Equations and Inequalities Problem Solving

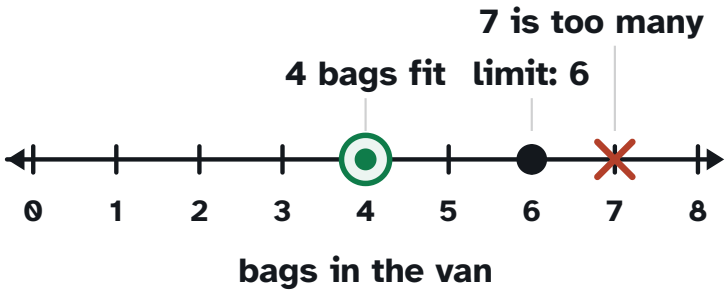
Decide if a situation gives one exact value or a limit, then pick an equation or an inequality.

Name _____ **Date** _____

Tonight's spark

Two notes are on the community-center board. A: a reward split equally among 5 people gave each $40. B: the van held no more than 6 bags. Which note gives ONE exact number?

Guess first: is it note A or note B?



Our class model: exact value or limit

Try it

Work it out on paper. Write one answer on each line.

1 **Which clue words usually point to an equation?**

A. split equally, each, is **B.** at most, at least **C.** no more than **D.** fewer than

My answer _____

2 **Which situation needs an inequality?**

A. A rope is exactly 9 feet. **B.** A box can hold at most 12 books.

C. Four friends share $20 equally. **D.** A book costs $7.

My answer _____

3 **Which situation needs an equation?**

A. A bag holds no more than 10 pounds. **B.** You need at least 15 points.

C. Six friends share 24 stickers equally. **D.** A ride allows at most 4 people.

My answer _____

4 **'Tickets cost $4 each and the total is $28.' Which model fits, with t tickets?**

A. $4t < 28$ **B.** $4t = 28$ **C.** $4t \geq 28$ **D.** $t - 4 = 28$

My answer _____

5 **'You can spend at most $30 on 3 gifts that cost g dollars each.' Which model fits?**

A. $3g \geq 30$ **B.** $3g = 30$ **C.** $3g \leq 30$ **D.** $g \div 3 \leq 30$

My answer _____

Talk about it

Which pile has a limit, and which has one exact value? How do you know?

Home mission

Find two notes at home: one with an exact amount ("the total is $28") and one with a limit ("at most 3 snacks"). Ask your student which is an equation and which is an inequality, and which words decided it.

If we get stuck, ask

- Does this give one exact answer, or a limit?
- Which words in the problem told you that?
- Could more than one number work here?

Do it on a screen

Scan or type the link. Same lesson, with hints.

eduwonderlab.com/lessons/8-7-part2/homework.html

How did it feel? (optional)

Tricky Getting there Got it